

Caching and stale information

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Data retention and deletion

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Untrusted input boundaries

Without opening the old lesson, explain one key idea from Untrusted input boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Approval gates and idempotency

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Caching and stale information. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Caching and stale information. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

TTL10 minutes; ages9, 10, 11. Corpus version changes at age9. Decide validity.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A cached result retains the limitations of its original evidence and can become stale independently of TTL. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

TTL10 minutes; ages9, 10, 11. Corpus version changes at age9. Decide validity.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled caching and stale information evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of caching and stale information. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Caching and stale information

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

Version 4 says 09:00; version 5 says 09:30. Cache key contains room and source version.

Caching and stale information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read authority

The current approved source version is 5.



2. Inspect cached key

The 09:00 entry belongs to version 4.



3. Fetch current evidence

Retrieve and cache version 5's 09:30 under its own key.

CHANGE ONE THING

The next request still uses approved version 5.

Explain the changed result using the matching lesson.

Identify the version mismatch in the cached 09:00 answer. What should the first version-5 request return, and what permits a later version-5 cache hit?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Retrieve and cache version 5's 09:30 under its own key.

1. Read authority
2. Fetch current evidence
3. Inspect cached key

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Fetch current evidence Retrieve and cache version 5's 09:30 under its own key. A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

Practice 2: explain the deciding evidence

Changed-case practice: The next request still uses approved version 5. Which conclusion is supported by the supplied facts?

1. The matching version-5 entry can supply 09:30 under the freshness policy.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The matching version-5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

Repair a missing building block

Review Data retention and deletion (ai-6/20). Goal: Explain the mechanism and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review Untrusted input boundaries (ai-6/18). Goal: Explain the mechanism and terminology of untrusted input boundaries. Short practice: Classify h3, H3, h30 and the trailing-space input by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Approval gates and idempotency (ai-6/15). Goal: Explain the mechanism and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
